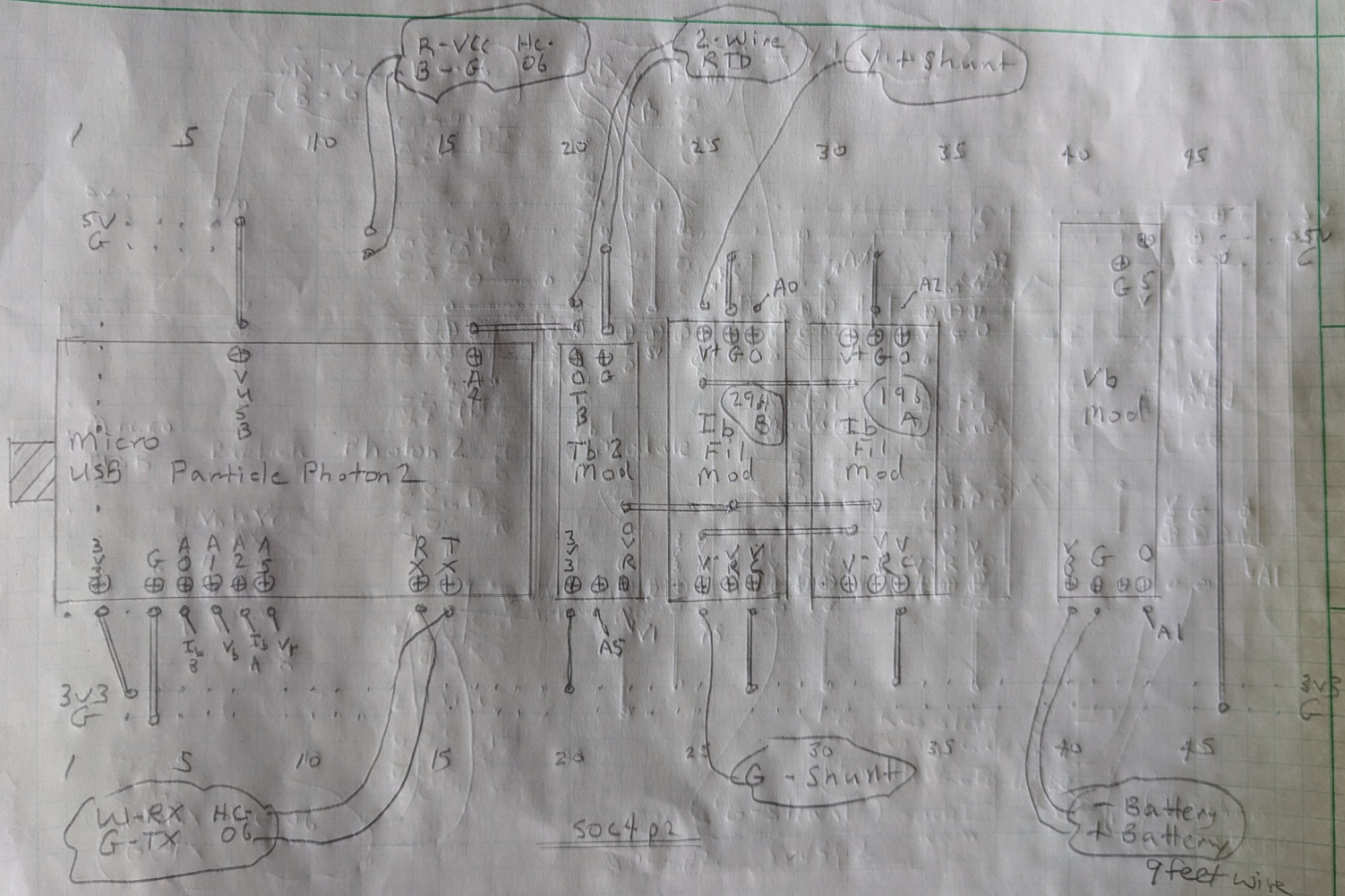
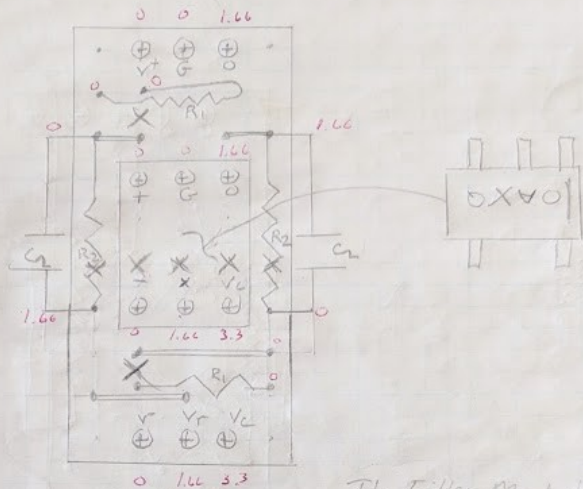






Ib Filter Module

- (1) 5x10 VeroBoard
- (1) SOT23-6 Board
- (1) OPA333AIDBVR (OAXO)

R_1 Ω	R_2 Ω	C_2 nF	#	Range xCS	Gain V/V	I s
5k1	75k	2.2	225	1.5	14.7	0.165
5k1	100k	0.15	154	0.11	14.6	0.150
5k7	150k	1.00	105	0.75	29.4	0.150
1k5	33k	4.7	475	1.0	22.0	0.155
1k5	330k	0.77	474	0.10	22.0	0.155

```

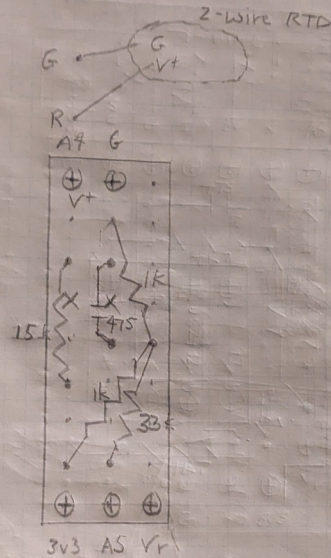
#define SHUNT_AMP_R1 (A)
#define SHUNT_AMP_R2
#define SHUNT_NOA_R1 (B)
#define SHUNT_NOA_R2
#define IB_ABS_MAX_AMP
#define IB_ABS_MAX_NOA
#define HOWE_IB_HI_LO *
  
```

0.1% Resistors
Ceramic Caps
#define HOWE_IB_HI_LO
 $R = 0.150 \pm 10\% \rightarrow R_2 C_2$
Gain = R_2 / R_1

1/2/2026

DAC

G-shunt ratings

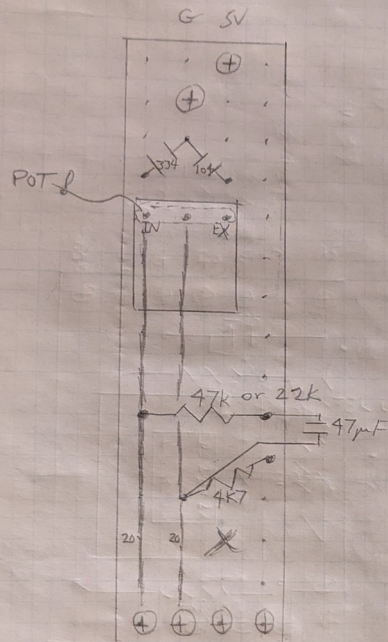


- (1) 15k (1%) RTD splitter
- (2) 1k (1%) Vr splitter
- (1) 33k (1%) LPF for Vr sensing
- (1) 475 Ceramic (4.7μF) LPF for Vr Sensing
- (1) 3 x 9 Vero Board

Tb2 Module

$$33k \times 4.7e-6 = 0.155s \text{ LPF}$$

DAG
6/9/2024



$$\tau_{24} = \frac{47k \times 4k7 \times 47e-6}{47k + 4k7} = 0.20 \text{ s}$$

$$\tau_{12} = \frac{22k \times 4k7 \times 47e-6}{22k + 4k7} = 0.18 \text{ s}$$

(2) 20 awg jumper Vb G x AI

(1) 334 Ceramic Cap 0.33 µF

(1) 104 Ceramic Cap 0.1 µF

(1) Electrolytic Cap 47 µF

(1) 47k 1% for 24V = V_b or 22k 1% for 12V = V_b

(1) 4k7 1%

(1) Taidacent LM7805 Eg Buck Converter

> 0.15 s Filtering Required (0.15 < τ < 0.20)

(1) 1.5 x 15 VeroBoard

(1) Liquid Tape (POT)

Vb module

DAG
5/17/2024

